RAHIL SHARMA

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SUMMARY

AI Platform and Data Engineer, recent MSc Computer Science graduate with 5+ years of combined industry, research, and teaching experience, including 2.5 years working on enterprise data integration and production data systems. My background spans data engineering, distributed systems, and machine learning. Experienced in Python, SQL, Power BI, cloud data platforms, ETL/ELT, data quality, automation, monitoring, and production operations. Skilled in building reliable data pipelines, investigating complex data issues, and translating operational requirements into practical solutions for technical and non-technical stakeholders. Particularly interested in interdisciplinary problems where reliable technical systems, human judgement, and business decision-making come together. Strong interest in financial markets, risk management, algorithmic trading, and using data to improve financial decision-making. Based in the Netherlands and available to start immediately.

RELEVANT WORK EXPERIENCE

Precisely, *Mumbai, IN*

Software Support Engineer I: Data Integration *Jan 2022 – Aug 2024*

- Built, configured, and maintained enterprise batch and real-time data pipelines using Precisely Connect CDC, Connect ETL, and Connect for Big Data across cloud, distributed, and legacy environments.
- Onboarded enterprise customers from proof of concept through production go-live, validating connectivity, source-to-target mappings, transformation logic, deployment readiness, and operational reliability.
- Developed mappings, schema configurations, and transformation logic using SQL, dbt, and Precisely DTL, with reconciliation controls for missing records, duplicate data, schema mismatches, and transformation errors.
- Built reusable integration patterns across relational databases, Kafka, Spark, Databricks, Snowflake, and cloud platforms, enabling downstream analytics and reporting teams to consume timely and reliable data.
- Monitored business-critical pipelines and investigated failures, replication latency, data inconsistencies, and performance issues before they affected downstream users.
- Used SQL, system logs, validation checks, and source-to-target reconciliation to identify root causes and verify the accuracy and completeness of production data.
- Worked directly with technical and non-technical stakeholders to understand operational requirements, explain data issues, and translate business workflows into practical data solutions.
- Identified opportunities to automate manual processes, introduce new data use cases, and streamline inefficient integration and monitoring workflows.
- Supported platform upgrades, migrations, performance tuning, testing, and controlled production releases through GitHub, Azure DevOps, and established CI/CD pipelines.
- Managed engineering and production issues through Jira, maintaining clear investigation histories, technical documentation, and resolution tracking.
- Created knowledge-base articles and shared reusable troubleshooting practices, improving knowledge transfer across the support engineering team.
- Contributed product and domain expertise to an internal AI/LLM pilot for support engineers, helping shape retrieval workflows and practical use cases around recurring production issues and technical knowledge.

Centrum Wiskunde & Informatica (CWI), Amsterdam, NL

Research Assistant: CSY Group (Security and Machine Learning) *Feb 2025 – Jun 2026*

- Built a distributed Python and gRPC system implementing privacy-preserving protocols, including encrypted communication, persistent storage, and configurable execution workflows.
- Translated theoretical specifications into a working software system, resolving ambiguities between formal designs and practical implementation constraints.
- Developed reproducible data-processing and benchmarking pipelines to evaluate correctness, scalability, performance, and deployment trade-offs across system configurations.
- Implemented systematic testing and validation workflows to identify failures, verify outputs, and maintain consistency across experiments.
- Collaborated with researchers and supervisors on architecture decisions, code reviews, experimental design, and technical documentation.

Vrije Universiteit, Amsterdam, NL

Senior Teaching Assistant *Dec 2024 – Jun 2026*

- Mentored students across machine learning, robotics, and systems engineering projects, providing technical guidance on architecture, implementation, and evaluation methodologies.
- Guided teams in debugging complex software systems, experimental design, and interpreting empirical results from data-driven systems.
- Facilitated technical discussions and supervised project development, helping teams translate theoretical concepts into working software solutions.
- Designed assessment frameworks and evaluation criteria to ensure reproducibility, technical rigor, and effective communication of system behaviour.

EDUCATION

***Master of Science (MSc) Computer Science (Big Data Engineering)**, Joint Degree* — *Sept 2024 – Jun 2026*

Vrije Universiteit Amsterdam (VU) & University of Amsterdam (UvA)

GPA: 8.5/10

Executive PG Program in Software Development with Specialization in Full Stack Development — *Aug 2022 – Oct 2023*

International Institute of Information Technology*, India*

***GPA:** 3.3/4*

Bachelor of Technology (B. Tech), Electronics and Telecommunication Engineering — *Aug 2018 – Jun 2022*

Symbiosis Institute of Technology, *Symbiosis International University, India*

***Percentage:** 74%*

SKILLS

Programming & Software Engineering: Python, SQL, Java, JavaScript (Node.js), REST APIs, gRPC, Unit Testing, FastAPI, React, Streamlit

Data Engineering: ETL/ELT, Change Data Capture, Batch and Real-Time Pipelines, Data Modelling, Source-to-Target Mapping, Data Quality, Reconciliation, dbt, Apache Spark, PySpark, Kafka, Power BI, Databricks, Snowflake, Hadoop, Hive, HDFS

Cloud & Infrastructure: Microsoft Azure, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Databricks, AWS, GCP, BigQuery

Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, Supervised Learning, Fraud Detection, Anomaly Detection, Model Evaluation

Databases & Enterprise Systems: PostgreSQL, MySQL, MS SQL Server, Oracle, IBM i / AS400, SQLite

AI & LLM Systems: RAG Architectures, Vector Retrieval, Semantic Chunking, Embedding Pipelines, Prompt Management, Multi-Model Evaluation, Model Routing, Automated AI Evaluation

MLOps & ML Platforms: Feature Stores, Online/Offline Feature Pipelines, Model Training and Deployment, Model Registry, Experiment Tracking, Model Monitoring, Drift Detection, Batch and Real-Time Inference, Reproducible ML Pipelines

Engineering Practices: Git, GitHub, Azure DevOps, CI/CD Workflows, Jira, Testing, Data Validation, Monitoring, Production Support, Root-Cause Analysis, Technical Documentation

PROJECTS

Selected project details, demos, and code links: [Portfolio Projects]

Multi-Model RAG & AI Evaluation Platform — VU Amsterdam | 2026

- Built an end-to-end AI platform using Python, FastAPI, React, semantic retrieval, and GPT, LLaMA, and Qwen model families.
- Developed reusable ingestion, retrieval, model-interaction, evaluation, and experiment-logging components to support comparisons across models and configurations.
- Designed APIs and a user-facing application that connected data preparation, retrieval, generation, evaluation, and observability into a unified workflow.
- Applied structured testing and evaluation to investigate response quality, retrieval performance, latency, and model behaviour.

Explainable AI Framework for Privacy-Aware Feature Attribution — UvA Amsterdam | 2025

- Developed an automated framework for **privacy-aware feature selection** aligned with **GDPR data minimization principles**.
- Combined **SHAP values and knowledge graphs** to provide interpretable and context-aware feature attribution.
- Built end-to-end pipelines supporting heterogeneous datasets and explainability workflows **across numerical and categorical features**.
- Integrated privacy-preserving representation learning mechanisms to support **secure and responsible AI applications**.

Auditable Financial Fraud Detection: Graph Features & Agentic Investigation — UvA Amsterdam | 2026

- Built and evaluated an end-to-end fraud detection pipeline on 6.3M+ financial transactions, with **a focus on reducing manual review** while keeping decisions explainable and auditable.
- **Engineered graph-based and anomaly-detection features** that improved detection on difficult, low-confidence cases, and separately validated the graph-based approach through a controlled fraud-ring experiment.
- Developed an **AI-assisted investigation workflow** that combined **model explanations, transaction context, and similar historical cases** into structured case rationales.
- Designed a **governance framework with a confidence-based escalation policy** that automatically resolved clear cases and routed conflicting decisions to human review.

OBFI: Implementation, Empirical Analysis, and Crossover Characterization of Competing Protocols — CWI Amsterdam | 2026

- Built a **distributed client–server system** in Python (gRPC, encryption, persistence) to implement and compare two cryptographic protocols.
- Translated formal theoretical protocol specifications into executable systems, resolving ambiguities in prior research.
- Designed and ran experiments across varying batch sizes and filter capacities to evaluate real-world system behavior for a sensitivity study.
- Developed a decision framework for selecting between protocols based on **deployment parameters and observed performance trends**.
- Ensured correctness and consistency through systematic testing across configurations.

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

- **Awards**: Winner AGBI Health-Tech Grand Challenge (NITI Aayog, 2022); Top-2 Finalist, TechBharat National Hackathon (2021); Precisely Recognition: Best Employee Award , Spot Award, Joy Giver Award
- **Certifications**: AWS Solutions Architect Associate, Azure Data Engineer Associate, Snowflake SnowPro Core, Databricks Lakehouse Data Engineer, Confluent Kafka Developer, Google Cloud AI Essentials, GitHub Foundations (GH-900), GitHub Copilot (GH-300)
- **Languages**: English (Fluent), Dutch (A2), French (A2)